

Tax Productivity and Development: A Quantitative Macroeconomic Analysis

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The Importance of Tax Collection

- Core task of government; prerequisite for any public spending
- Central to 'state capacity' theory of development
 - Besley-Perrson, 2009, 2010, 2013; Dincecco-Onorato 2018; Gennaioli-Voth 2015; Gordon-Li, 2009; Tilly 1990; others

Challenges Faced in the Literature

- Tax collection capacity or not easy to measure
 - Typically proxied by Taxes/GDP but this is quite indirect
 - Theories of state capacity $\Rightarrow$ tax capacity increases Taxes and GDP
- Little consensus on *quantitative importance* of tax collection capacity
 - Few answers to: How much would GDP / capita increase in the long-run if we doubled Africa's tax collection capacity?

Overview of This Paper

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 - Tax collections / collection expenditures
 - Document strongly positive correlation with GDP / capita in cross-section and country time series

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- Dynamic G.E. model to interpret data + simulate counterfactuals
 - Ramsey model with tax administration sector, modern sector, and traditional sector
 - Production function of tax administration: $\tau = G(A^C, L_C)$
 - A^C = tax productivity, L_C = number of collectors, τ tax rate on modern sector wage bill,
 - Public capital and private capital complements in production a la Baxter-King (1993)
 - Lower collection capacity $\Rightarrow$ lower public capital $\Rightarrow$ lower GDP per capita

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- (Goal) Infer differences in tax collection capacity across countries
- Counterfactually doubling Sub-Saharan Africa's tax collection capacity would raise GDP / capita in the long run by $\sim X\%$
 - Order of magnitude larger than simple "Hulten" logic, which suggests $Y\%$

Empirical Findings: Tax Collections / Expenditures

Measuring Tax Collections/ Expenditure

- Unit of Observation: tax administration

Measuring Tax Collections/ Expenditure

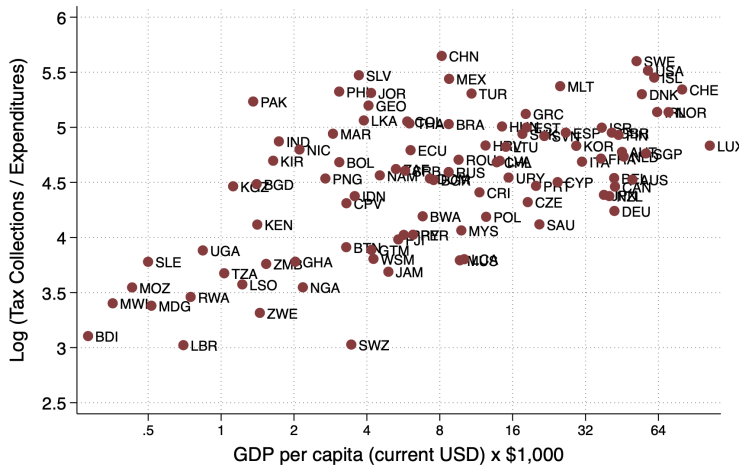
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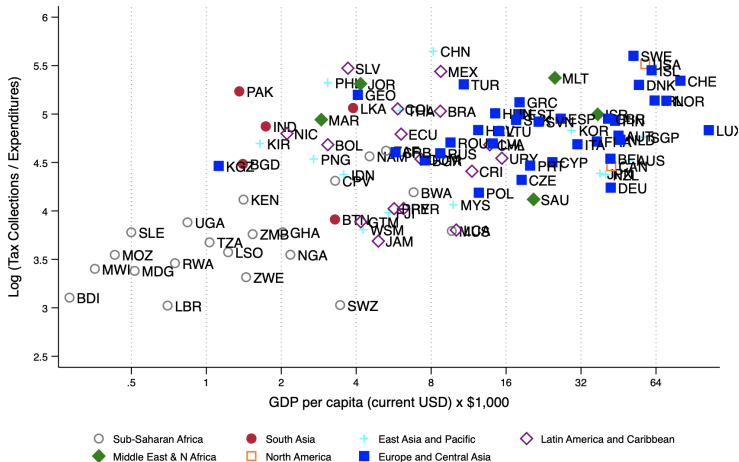
Measuring Tax Collection / Expenditure

- Data Source
 - National tax authority documents
 - IMF Revenue Administration Fiscal Information Tool
- Unit of Observation: tax administration
- 98 countries, 625 country-year observations
- T = Tax Collections including personal income and corporate taxes, VAT, sales and custom taxes
- X = Official tax collection authority budget
- Tax collection over collection expenditure, T/X

Tax Collection / Expenditure Increases with GDP per capita



Tax Collection / Expenditure Increases with GDP per Capita

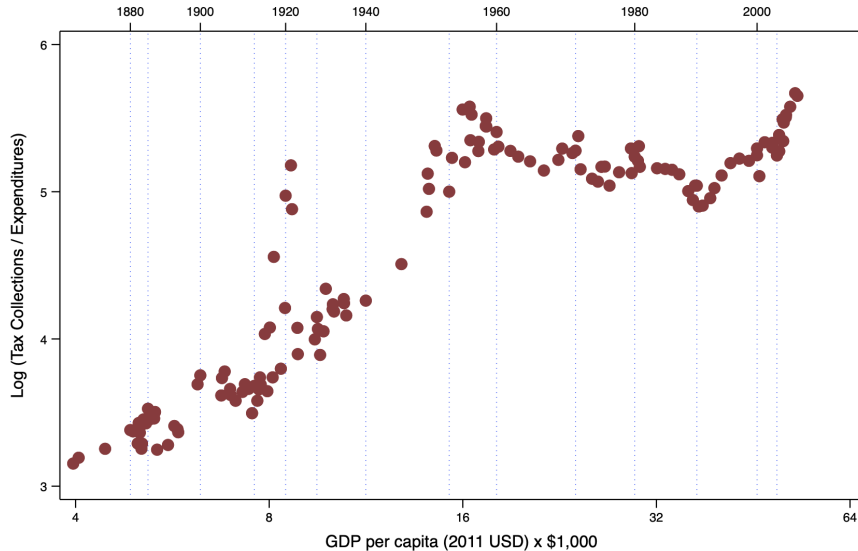


Descriptive Regressions: Tax Collection / Expenditure & GDP Per Capita

	(1) Log (T/X)	(2) Log (T/X) Exclude if not responsible for VAT/sales taxes	(3) Log (T/X) Exclude if not responsible for customs duties	(4) Log (T/X) Exclude if responsible for lotteries	(5) Log (T/X) Exclude if responsible for social security contributions
Log GDP pc	0.256*** (0.0363)	0.257*** (0.0367)	0.268*** (0.0367)	0.258*** (0.0362)	0.222*** (0.0443)
Countries	98	95	88	96	72
R-squared	0.341	0.346	0.384	0.351	0.264

Note: Each column represents one regression of log tax productivity, defined as taxes collected divided by tax administration expenditures, on log GDP per capita. *** means significant at 1%, ** means 5%, and * means 10%

Tax Collection / Expenditure in the US: 1867 to 2017



Descriptive Regressions: Historical Time Series

	(1) Log (T/X)	(2) Log (T/X) Exclude if not responsible for customs duties	(3) Log (T/X) Exclude if responsible for social security contributions	(4) Log (T/X) Exclude if changes in responsibilities
Log Real GDP pc	0.637*** (0.175)	0.636*** (0.197)	0.684*** (0.163)	0.644*** (0.193)
Observations	625	546	519	578
R-squared	0.563	0.541	0.574	0.541
Country FE	yes	yes	yes	yes

Note: Each column represents one regression of log tax productivity on on log GDP per capita with country fixed effects. the sample consists of 6 OECD countries, 1 from Latin America, 4 from Asia, and 5 from Sub-Saharan Africa. *** means significant at 1%, ** means 5%, and * means 10%

Model of Tax Productivity and Development

Overview of Model

- Positive correlation between collections / expenditure and GDP / capita suggestive of higher tax capacity in rich countries, but not directly interpretable
- Need structural model to:
 - Formalize notion of **tax productivity**
 - Measure collection capacity in cross section of countries
 - Simulate effects of counterfactually raising collection capacity

Overview of Model Environment

- Ramsey model à la Chamley (1986)
- Public and private capital in the production function (Baxter-King, 1993)
- Modern and Traditional production sectors
- Novelty: Tax administration sector
 - One input: tax collectors
 - Output: tax rate on modern sector wages
 - Tax productivity is exogenous, captures “tax collection capacity”

Technology: Modern and Traditional Sectors

- Modern firms production function:

$$Y_M = F(A^M, k_G, k, L_M),$$

- k_G : public capital,

Firms hire workers at wage $w_{M,t}$ and rent capital at rate r_t

$$\max_{k_t, L_{M,t}} F(A^M, k_{G,t}, k_t, L_{M,t}) - w_{M,t} L_{M,t} - r_t k_t.$$

- **Traditional Sector:**

Firms in the traditional sector have a linear production function:

$$Y_T = A^T L_T,$$

Workers are hired at a fixed wage $w_T = A^T$.

Public Sector and Government

Tax Collection:

Taxes T_t are collected on modern sectors wages at rate $G(A^C, L_{C,t}) \leq 1$,

$$T_t = G(A^C, L_{C,t}) w_{M,t} L_{M,t}.$$

Government budget: can borrow to finance budget.

$$w_{M,t} L_{C,t} + R_t b_t + k_{t+1}^G \leq T_t + b_{t+1} + (1 - \delta^G) k_t^G.$$

Public capital evolves as:

$$k_{G,t+1} = (1 - \delta^G) k_{G,t} + T_t.$$

Government cannot borrow to finance public capital.

Households

Modern and traditional sector workers form a representative household that owns all capital and supplies total labor. The household maximizes its lifetime utility:

$$\max_{\{c_t, k_{t+1}, b_{t+1}\}} \sum_{t=0}^{\infty} \beta^t u(c_t, L_{M,t} + L_{T,t}),$$

$$\text{subject to: } c_t + k_{t+1} + b_{t+1} \leq (1 + r_t - \delta)k_t + (1 - G(A^C, L_{C,t}))w_{M,t}L_{M,t} \\ + w_T L_{T,t} + R_t b_t.$$

The initial private capital is given by $k_0 = \bar{k}_0$. Here, r_t is the return on private capital, δ the depreciation rate, and R_t the return on bonds.

Equilibrium

A competitive equilibrium in this economy consists of allocations

$$\{c_t, L_{T,t}, L_{M,t}, L_{C,t}, k_t, k_{G,t}, b_t\}$$

and prices

$$\{w_{T,t}, w_{M,t}, r_t, R_t\},$$

such that:

1. Firm Optimality:

- Modern firms take prices and $k_{G,t}$ as given, solve (17), and earn zero profit.
- Traditional firms hire labor at wage $w_T = A^T$.

2. Household Optimality: The household solves (??) for given prices and tax collection personnel.

3. Government Conditions: The government budget constraint (18) and public capital accumulation (18) hold.

4. Market Clearing: The goods market clears:

$$c_t + k_{t+1} + w_{M,t}L_{C,t} + k_{G,t+1} \leq (1 + r_t - \delta)k_t + w_{M,t}L_{M,t} + A^T L_{T,t} + (1 - \delta^G)k_{G,t}.$$

Planner's Problem Setup

To study optimal policy, the planner (or tax authority) chooses the sequence of tax collection personnel $\pi = \{L_{C,t}\}$ to maximize the household's lifetime welfare:

$$\max_{\pi} \sum_{t=0}^{\infty} \beta^t u(c_t(\pi), L_{M,t}(\pi) + L_{T,t}(\pi)),$$

subject to:

- the economy's resource constraints,
- an **implementability condition** that ensures allocations are consistent with household optimization, and
- the government's budget constraint.

Lemma

The implementability condition on the planner's allocation choice is:

$$\sum_{t=0}^{\infty} \beta^t \left[u_c(t) c_t - u_{L_M}(t) L_{M,t} - u_{L_T}(t) L_{T,t} \right] = u_c(0) R_0 k_0.$$

Optimal Tax Collection

At an interior steady state, the optimal tax collection choice satisfies:

$$G_{L_C}(A^C, L_C) = (1 - (1 - \delta_G)\beta) \frac{1}{L_M} \frac{F_k + 1 - \delta}{(F_G - \delta_G) - (F_k - \delta)}.$$

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Economic Interpretation:

- A high F_k (marginal return on private capital) favors private investment over public hiring.
- A high F_G (marginal return on public capital) enhances the attractiveness of public sector spending.
- A higher tax collection productivity A^C improves the effectiveness of tax collectors.
- A larger modern sector (L_M) expands the taxable base.

Transition Dynamics

Lemma

Assume that household preferences are separable in consumption and labor, with CRRA utility in consumption, and that the tax production function is linear in personnel $G(A^C, L_C) = A^C L_C$. Then an interior steady state exists and the optimal personnel choice over the transition (for $t > 0$) satisfies:

$$\frac{(F_G(t+1) - \delta_G) - (F_k(t+1) - \delta)}{F_k(t+1) + 1 - \delta} = \frac{1}{A^C} \left(\frac{1}{L_{M,t}} - \frac{\beta(1 - \delta_G)}{L_{M,t+1}} \right).$$

Quantitative Analysis

Quantitative Analysis: Setting

We introduce capital wedges $\mu_{k,t}$ in the household's capital decision to capture realistic distortions in emerging markets and the potential distortionary effects of public capital. These wedges modify the standard Euler equation:

$$\frac{\beta u_c(t+1)}{u_c(t)} = \frac{1}{(1 - \mu_{k,t+1})(1 + r_{t+1} - \delta)}.$$

Extended Euler Equation and Public Capital

Proof Outline:

- The modified first-order condition (FOC) for private capital incorporates the wedge $\mu_{k,t+1}$ as follows:

$$1 = \frac{1}{(1 - \mu_{k,t+1})(1 + r_{t+1} - \delta)} \left[\dots \right].$$

- A corresponding FOC for public capital is derived, with similar adjustments.
- In steady state, the relevant ratios simplify, leading to expressions such as:

$$\frac{1 - \beta (1 - \delta_G + G(A^C, L_C) F_{L_M} G L_M)}{G_{L_C}(A^C, L_C) L_M} = \beta \left[F_G + (1 - \delta_G) - F_{L_M} G L_C \right] - 1.$$

Functional Forms and the Steady State

We set the following functional forms:

$$G(A^C, L_C) = A^C L_C^\eta,$$

$$F_M(k^G, k, L_M) = A_M (k^G)^\zeta k^\mu L_M^{1-\mu},$$

$$F_T(L_T) = A_T L_T,$$

$$u(c) = \frac{c^{1-\gamma}}{1-\gamma},$$

$$u(L_M + L_T) = \frac{(L_M + L_T)^{1/\epsilon}}{1 + 1/\epsilon}.$$

A steady state is determined by the following system of 9 equations in the 9 unknowns ($w_M, w_T, r, L_M, L_T, L_C, k, k_G, c$):

(i) Optimal collection:
$$\frac{1 - \beta(1 - \delta_G)}{G_{L_C}(A^C, L_C) L_M} = \beta \left[F_G + (1 - \delta_G) \right] - 1 + \beta \left(\frac{1 - \eta}{\eta} \right) F_{L_M G} L_C,$$

(ii) Private capital:
$$1 = \beta \left(F_k + 1 - \delta \right) + \beta \frac{1 - \eta}{\eta} F_{L_M, k} L_C,$$

(iii) Public capital:
$$k_G = \frac{1}{\delta} G(A^C, L_C) w_M L_M.$$

Conclusion

We have presented a model that integrates:

- A dual-sector economy with heterogeneous workers,
- A framework where tax revenue finances public capital,
- A planner's problem that internalizes household choices via an implementability condition, and
- A quantitative extension that incorporates capital distortions.

This framework allows for the analysis of trade-offs between private capital accumulation and public spending on tax collection, as well as the study of steady state and transitional dynamics in emerging markets.

Households

- Representative household with measure 1 of individuals; log utility
- The distribution of human capital is given by a c.d.f. $G(h)$ and p.d.f. $f(h)$
- Simplifying assumption I (for now): individuals pool income, consumption
- Simplifying assumption II (for now): inelastic labor supply

Quantitative Analysis

Quantitative Analysis

- Goal:
 - Infer tax collection capacity (Z) across countries
 - Simulate long-run aggregate effects of raising collection capacity

Experiment: Double Tax Collection Capacity (Z) of Ghana



Conclusion and Future Work

- New measures of tax collection capacity across countries
- Richest countries 30 more collection capacity than poorest
- Disciplined by new data of tax collection per expenditure on collection
- Long-run effects of raising capacity could be substantial – much larger than those suggested by simple Hulten calculation
- Future work: include tax avoidance behavior, consumption taxes, labor supply distortions, self-employment sector, heterogeneous cost of taxation by sector, alternative elasticities of substitution between public and private capital